Dang Phuong Nam

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Education

FPT University — Hanoi, Vietnam
Bachelor of Artificial Intelligence — *Oct 2023 – Dec 2026 (Expected)*

Publications

Conflict-Aware RAG Routing: Balancing Cost and Accuracy via Context Contradiction 2026

- **Dang Phuong Nam**, Tran Huy Tuan, Doan Duy Long, Cao Van Mai. *13th IEEE International Conference on Intelligent Systems (IEEE IS'26).* **First author**; accepted & presented; proceedings to appear.

Probabilistic Modeling of Stock Breakout Success during Consolidation Phases in the Vietnamese Equity Market 2026

- Doan Duy Long, Tran Huy Tuan, **Dang Phuong Nam**, Cao Van Mai. *11th International Conference on Information Technology and Digital Applications (ICITDA 2026).* Accepted.

Experience

Full-stack Developer Intern — Aug 2025 – Dec 2025
XTEK — *Hanoi, Vietnam*

- Learned full-stack web development through mentored internship projects using React.js and Spring Boot
- Built a shoe e-commerce frontend with React.js and a student management backend with Spring Boot

Projects

Conflict-Aware RAG Routing | *First Author* | *PyTorch, Hugging Face, scikit-learn* — GitHub | 2026

- Designed a bidirectional NLI conflict score and a 5-feature router (local SLM vs. DeepSeek API)
- Built leakage-resistant nested 5-fold budget calibration; evaluated on 3,000 multi-hop questions
- Saved ∼**35% of API calls** while keeping **83.2–92.2% of Always-LLM F1**; **+14.35 F1** on high-conflict cases

Multi-Task Stock Transformer | *Capstone, Vice Team Leader* | *PyTorch, Qwen2.5, PatchTST* — 2026 – Present

- Co-designing a decoder-only SLM (Qwen2.5-0.5B) fusing price patches, financial text and ESG for trend prediction
- Proposed an NLI-based consistency loss that penalizes rationales contradicting the predicted trend

VietSign Edge AI | *Team Leader* | *PyTorch, YOLOv8n, MobileNetV3, TFLite* — GitHub | 2026

- Led a team building a fully offline Android pipeline: YOLOv8n sign detection followed by MobileNetV3-Small classification of 48 Vietnamese traffic sign classes
- Owned the data and classifier pipeline: knowledge distillation from ResNet-50, ∼50% L1 pruning and INT8 QAT, shrinking the classifier to 1.88 MB (5.07 MB for both models)
- Achieved **82.3% mAP@0.5** (INT8 detector) and **91.9% Top-1** (classifier) on held-out test sets

Luc Bat Poem Generator | *Solo Project* | *Qwen3-8B, Hugging Face, QLoRA* — GitHub | 2026

- Built a neural-symbolic generator for Vietnamese regulated poetry: a custom logits processor enforces syllable and line constraints with soft tone/rhyme penalties, plus a verifier and N-best reranker
- Raised strictly rule-valid poems from **0/12** (raw Qwen3-8B) to **12/12** on held-out prompts at ∼0.84 ms/step overhead; 195 regression tests

Technical Skills & Certifications

Languages: Python, Java, TypeScript / JavaScript, SQL
ML & Deep Learning: PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, QLoRA / PEFT
LLM & RAG: Hybrid retrieval (BM25 + dense, RRF), NLI cross-encoders, constrained decoding, LLM APIs
Computer Vision & Edge: YOLOv8, MobileNetV3, OpenCV, TFLite, knowledge distillation, pruning, INT8 QAT
Backend & Tools: FastAPI, Spring Boot, React.js, PostgreSQL / pgvector, Docker, Git, Linux
Certifications: NLP (DeepLearning.AI), RL (U. of Alberta), Deep Learning (NVIDIA DLI), IBM AI Workflow